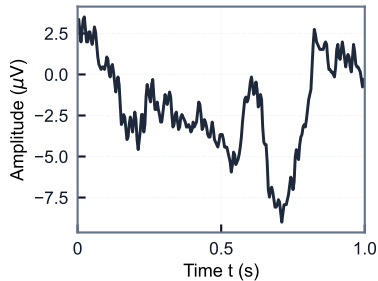
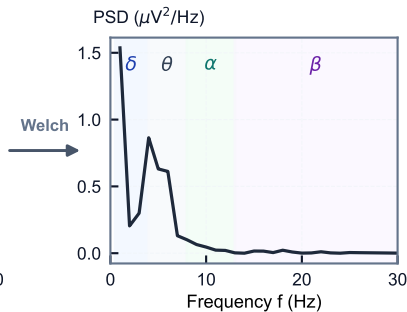
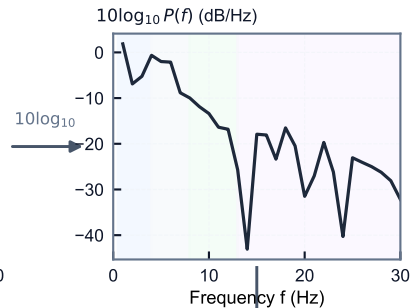
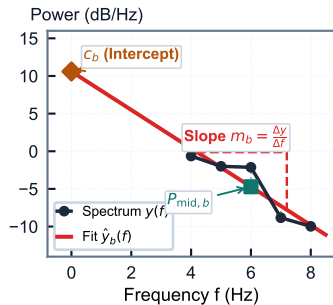


(a) Raw EEG Epoch $x(t)$ **(b) Welch PSD $P(f)$** **(c) Log-PSD Dynamics $y(f)$** **(d) Sub-Band OLS Fit $\hat{y}_b(f)$** **(e) Biomarker Vector $\mathbf{x}_{\text{slope}}$**

- Sub-Band Vector:**
 $\mathbf{f}_b = [m_b, C_b, P_{\text{mid},b}]^\top$
- Channel Vector:**
 $\mathbf{f}_c = [\mathbf{f}_\delta^\top, \mathbf{f}_\theta^\top, \mathbf{f}_\alpha^\top, \mathbf{f}_\beta^\top]^\top$
- Channel Dim:**
 $\dim(\mathbf{f}_c) = 4 \times 3 = 12\text{D}$
- Multichannel:**
 $\mathbf{x}_{\text{slope}} = [\mathbf{f}_1^\top, \dots, \mathbf{f}_C^\top]^\top$

$$\mathbf{x}_{\text{slope}} \in \mathbb{R}^{C \times 12}$$

$$(C = 18 \rightarrow 216\text{D})$$

Concatenate